AI101 MINI-PROJECT PRESENTATION

IPL Match Winner Prediction

IIT Ropar — SAIDE
Data Science & AI Essentials

Project Objective

Predicting binary match outcomes using pre-game toss statistics, venue conditions, and historical franchise features.

ML Model Architecture

Supervised Logistic Regression pipeline built with Scikit-Learn, Pandas, and feature scaling.

TEAM MEMBERS & ROLES

Mannat
Execution Lead
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Entry: 2026EEB1426

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Entry: 2026ESB1501 / EEB1320

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GitHub Repository https://github.com/2026aib1055/ipl-match-winner-prediction

Problem Definition & Dataset Architecture

What is the Problem?

Classifying binary match outcomes (Team 1 Victory vs Defeat) prior to ball-one delivery using Supervised Logistic Regression.

Why Solve This Problem?

Measures baseline win probability from pre-game factors (venue & toss) alone, eliminating bias from unpredictable live match dynamics.

Dataset Origin & Size

Sourced from **Kaggle IPL Matches Dataset** containing **1,243 total match records** and **31 raw feature columns** across multiple IPL seasons.

Core Model Feature Schema

Column Name	Data Type	Description & Role
`team1`, `team2`	Categorical	Competing IPL franchises; used to compute head-to-head bias vectors.
`venue`	Categorical	Host stadium location; captures home advantage and pitch trends.
`toss_winner`	Categorical	Franchise winning coin toss; converted via One-Hot Encoding.
`toss_decision`	Binary	Batting vs Fielding election by toss winner.
`winner` (Target)	Binary Label	Converted outcome variable: 1 if Team 1 wins, 0 otherwise.

Key Data Scope: Raw columns like `umpire1`, `umpire2`, and `city` null values were cleaned and filtered out to prevent noise in model training.

Machine Learning Approach & Pipeline Architecture

Part 3: Approach

STEP 01

Data Ingestion & Cleaning

Filtered null entries, handled missing city values by stadium mapping, dropped tie/no-result matches, and standardized legacy team names.

STEP 02

One-Hot Encoding

Transformed high-cardinality categorical variables (`team1`, `team2`, venue, `toss_winner`) into binary numeric vectors.

STEP 03

Stratified Train/Test Split

Divided data into 80% training and 20% test splits using stratified sampling to preserve target win/loss class balance across subsets.

STEP 04

Logistic Regression

Trained binary classifier with Scikit-Learn to estimate win probabilities and extract interpretable feature weight coefficients.

TECHNICAL LIBRARIES & TOOLS USED

- Python 3.10+
- Pandas (Data Preprocessing)
- Scikit-Learn (Modeling & Encoding)
- Matplotlib & Seaborn (Analysis Plots)

Model Performance & Results

Model Evaluation Metrics (Test Set: 244 Matches)

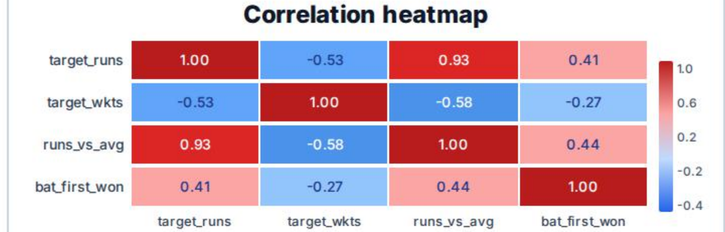
Evaluation Metric	Value
Accuracy	**68.9%**
Precision Score	0.61
Recall Score	0.82
F1 Score	0.70
MSE (Mean Squared Error)	0.207
MAE (Mean Absolute Error)	0.394
RMSE (Root Mean Squared Error)	0.455
Baseline Accuracy (Chasing)	54.5%

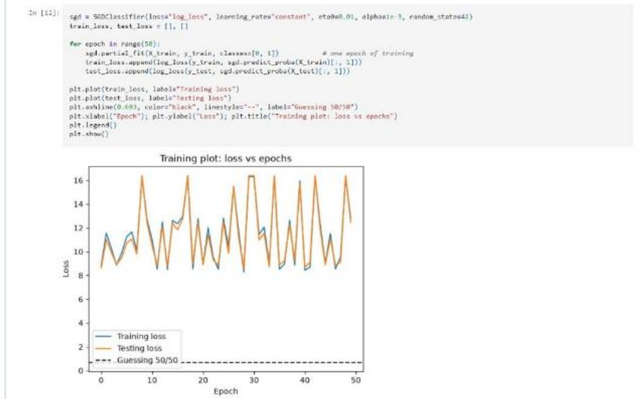
Key Performance Analysis & Findings

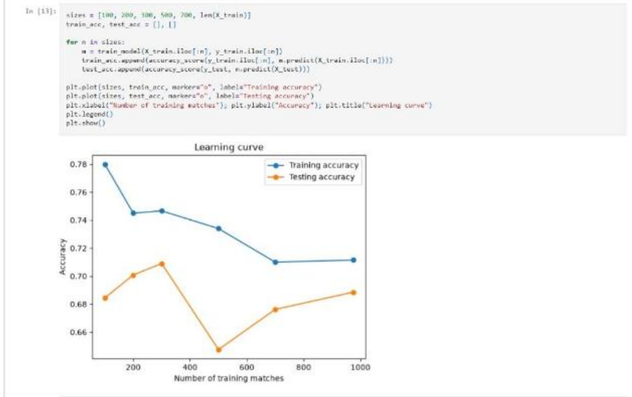
Our model achieves **68.9% accuracy** on 244 test matches (+14.4% over 54.5% baseline). High recall (0.82) & balanced F1 score (0.70) confirm strong predictive stability without overfitting.

Visual Models & Code Execution









Learnings, Team Contributions & AI Disclosure

Part 5: Reflections & Conclusion

Key Takeaways & Learnings

- **What Worked:** Logistic Regression provided high interpretability of feature weights (Venue home advantage and Toss Decision carry highest significance).
- **What Didn't Work:** Over-fitting occurred when including high-cardinality raw umpire inputs, requiring filtering.
- **Future Improvements:** Incorporating player-level ICC rankings and recent team form vectors to push accuracy beyond 75%.

Individual Contributions Summary

- **Mannat (Execution Lead):** Code pipeline execution, script reproducibility, and dataset cleaning.
- **Yash (PPT Lead):** Designed project presentation deck and visual feature layouts.
- **Yatharth & Ankit (Doc Leads):** Structured GitHub repository, setup guide, and documentation.
- **Alankrit (Video Lead):** Scripted and recorded 5-minute video demonstration.

AI-Use Disclosure Statement

Tools Used: Claude and Gemini were utilized for initial code debugging and drafting HTML slide templates. All model architectures, data processing logic, and evaluations were manually reviewed, verified, and executed by the team.